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BDACAP / Data Analytics with Capstone Project

Project 1 - Environment, APIs, Git, and Data Sourcing (15%)

Financial Data Gathering and Validation Summary Report

1. Introduction

The purpose of this stage was to collect historical financial stock market data for five publicly traded companies and evaluate its quality before conducting further financial analysis. Python, yfinance and pandas were used to retrieve, store and validate the datasets.

2. Data Collection

Historical stock market data was downloaded for the following companies:

- Apple Inc. (AAPL)
- Microsoft Corporation (MSFT)
- Tesla Inc. (TSLA)
- Amazon.com Inc. (AMZN)
- Alphabet Inc. (GOOG)

Each dataset contained 1,255 observations and included Date, Open, High, Low, Close, Adjusted Close, Volume, Dividends and Stock Splits.

The five individual CSV files were successfully saved in the project's data/raw/ directory.

Total observations collected: 6,275.

3. Data Quality Assessment

A Python validation script using pandas was developed to examine missing values, duplicate dates, invalid dates, unusual date gaps, invalid prices, invalid trading volumes and OHLC inconsistencies.

The validation results showed:

- Zero missing values.
- Zero duplicate dates.
- Zero invalid dates.

- Zero unusual date gaps.
- Zero invalid prices.
- Zero invalid trading volumes.
- Zero OHLC inconsistencies.

All five datasets contained 1,255 observations and passed the implemented validation checks.

4. AI-Assisted Validation

AI was used to independently critique the validation approach and recommend additional integrity checks. Recommendations included comparing selected closing prices against an independent financial data provider, checking chronological date ordering, identifying zero trading volumes, reviewing unusually large daily price movements and confirming consistent date coverage.

The AI assessment concluded that the datasets demonstrated strong initial data quality and were suitable for proceeding to exploratory analysis, subject to additional verification.

5. Limitations and Recommendations

Although the initial checks passed, successful validation does not guarantee that all downloaded figures are accurate or complete. Further verification should include independent price comparisons, exchange-calendar checks, zero-volume analysis and investigation of unusual market movements. Yahoo Finance data accessed through yfinance should also be treated as subject to provider availability and applicable usage terms.

6. Conclusion

The data gathering and initial validation stages were completed successfully. Five stock datasets containing 6,275 observations were collected and stored as individual CSV files. No errors were detected by the implemented quality checks.

The datasets provide a suitable starting point for the next stage of financial data analysis, with additional independent verification recommended before drawing investment or market conclusions.